

L5 v0.3 — Elie’s $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ absorbed (supersedes Keeper’s 281 single-form). Factor-2 cascade made explicit: substrate-predicted $\Lambda^{(1/4)} \approx 4.85$ meV vs observed 2.4 meV (factor 2.02); m_e formula at 0.73% requires observed Λ (SEARCH-FIT per Cal), at factor 2 if using substrate-predicted Λ . The factor-2 gap is the L5 mechanism work that remains. Honest disposition per Cal #182 brake on Keeper L5 framing.

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L5 v0.3 — Elie 280 absorbed; factor-2 cascade explicit

0. Per Cal #182 brake on Keeper framing + Elie 280 refinement

Cal #182 caught Keeper’s L5 framework headline overstatement (calling it “closure framework day” → reframed to “closure-ARCHITECTURE framework”). 5th symmetric audit-chain event Saturday. Per Cal’s brake + Elie’s refinement:

- Λ -exponent form: Elie’s $280 = 2^{\{N_c\}} \cdot n_C \cdot g = 8 \cdot 5 \cdot 7$ is 5-fold over-determined and cleaner than Keeper’s $281 = 2 \cdot N_{\max} + g$. v0.3 absorbs.
- My v0.2 m_e candidate: properly tier-marked as Tier 2 STRUCTURAL SEARCH-FIT (Cal’s tier) — not derivation.
- Factor-2 cascade: needs explicit acknowledgment.

1. Elie’s $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ (5-fold over-determined)

$\Lambda_{\text{BST}} = \exp(-280)$ where $280 = 2^{\{N_c\}} \cdot n_C \cdot g$ (substrate-natural product of $\{2^{\{N_c\}}, n_C, g\}$).

Verification: $2^3 \cdot 5 \cdot 7 = 8 \cdot 5 \cdot 7 = 280$ (cleaner than $281 = 2 \cdot N_{\max} + g$ which uses $N_{\max}$ in linear form).

5-fold over-determined per Elie: 280 has multiple substrate-primary decompositions all using only $\{2, N_c, n_C, g, C_2\}$: $- 2^{\{N_c\}} \cdot n_C \cdot g$ - (other equivalent forms via $N_c + n_C = 8 = 2^{\{N_c\}}$, etc.)

So 280 is the preferred substrate-natural Λ -exponent over 281.

2. $\alpha^{57} \approx \Lambda_{\text{BST}}$ with 280

With $\Lambda_{\text{BST}} = \exp(-280)$: $-\alpha^{57} = \exp(-57 \cdot \ln(137)) = \exp(-280.44)$. - $\Lambda_{\text{BST}} = \exp(-280)$. - Discrepancy: $\exp(-0.44) = 0.644$. - $\alpha^{57} / \Lambda_{\text{BST}} = 0.644$ (factor ~ 1.5).

Slightly closer than with 281 (factor was $0.57 = 1/1.75$). Still Tier 2 STRUCTURAL ($\sim 50\%$ in dimensionless), NOT exact.

3. The factor-2 cascade (explicit)

Substrate $\Lambda^{(1/4)}$ prediction from $\Lambda_{BST} = \exp(-280)$: - Λ_{BST} in Planck units = $\exp(-280) = 2.5 \times 10^{-122}$. - $\Lambda^{(1/4)}$ in Planck mass units = $\exp(-70) = 4.0 \times 10^{-31}$. - $\Lambda^{(1/4)}$ in eV = $4.0 \times 10^{-31} \times 1.22 \times 10^{28} \text{ eV} \approx 4.85 \text{ meV}$ (substrate prediction). - $\Lambda^{(1/4)}$ observed $\approx 2.4 \text{ meV}$. - Substrate/observed = 2.02 (factor ~ 2 off).

My v0.2 m_e formula with substrate-predicted $\Lambda^{(1/4)}$: - $m_e = (N_c/n_C) \cdot N_{\max}^4 \cdot \Lambda^{(1/4)}_{\text{substrate}} = 2.114 \times 10^8 \cdot 4.85 \times 10^{-3} \text{ eV} = 1.025 \text{ MeV}$. - Observed $m_e = 0.511 \text{ MeV}$. - Substrate/observed = 2.006 (factor ~ 2 off).

Factor-2 cascade: the substrate's Λ prediction is off by $\sim$ factor 2 from observed; my m_e formula's coefficient $(N_c/n_C) \cdot N_{\max}^4$ is calibrated to match observed $\Lambda^{(1/4)}$; when I use the substrate-predicted $\Lambda^{(1/4)}$ instead, m_e comes out at factor 2 of observed.

4. Honest disposition: SEARCH-FIT not derivation

Cal's tier marking applies: my v0.2 m_e formula is a Tier 2 STRUCTURAL SEARCH-FIT because: - I searched substrate-primary combinations for the ratio $m_e/\Lambda^{(1/4)} \approx 2.13 \times 10^8$. - Found $(N_c/n_C) \cdot N_{\max}^4 \approx 2.114 \times 10^8$ (matches at 0.73%). - The SEARCH found a substrate-natural form that fits OBSERVED $\Lambda^{(1/4)}$. - This is NOT a derivation from substrate mechanism (which would predict $\Lambda^{(1/4)}$ to match observed within precision).

The factor-2 cascade is the GAP between SEARCH-FIT and DERIVATION: - SEARCH-FIT: m_e formula matches observed Λ at 0.73%. - DERIVATION: substrate predicts Λ at factor 2 off + m_e formula coefficient.

Multi-week L5 mechanism work to close the gap: 1. Refine the $\alpha^{57} \approx \Lambda_{BST}$ relation (find what discrete-correction structure resolves the ~ 0.5 in log space). 2. Derive the $(N_c/n_C) \cdot N_{\max}^4$ coefficient from substrate Bergman kernel / commitment-density mechanism (per Keeper L5 framework Section 3). 3. Make $\Lambda^{(1/4)}_{\text{substrate}}$ predict observed at $< \text{few percent precision}$.

5. The honest L5 state

REFRAMED per Cal #182 (not "closed"): L5 has architecture (Keeper framework) + concrete CANDIDATE formula (Lyra v0.2 SEARCH-FIT) + open multi-week derivations (D1+D2+D3 deliverables).

Tier per Two-Tier + Cal SEARCH-FIT: - Architecture: FRAMEWORK (Keeper). - m_e formula at 0.73% (fitted to observed Λ): Tier 2 SEARCH-FIT (Cal). - $\alpha^{57} \approx \Lambda$: Tier 2 STRUCTURAL approximate. - Substrate Λ prediction: factor 2 off observed. - All open at multi-week mechanism work.

NOT promoted to operational criterion. The factor-2 cascade is the honest gap that the multi-week L5 derivations need to close.

6. Honest scope + tier

RIGOROUS (arithmetic): - Elie 280 = $2^{\{N_c\} \cdot n_C \cdot g} = 8 \cdot 5 \cdot 7$ (Tier 1 EXACT integer identity). - $\alpha^{57} = \exp(-280.44)$ computed. - $\Lambda_{BST} = \exp(-280) = 2.5 \times 10^{-122}$ (with substrate-natural integer 280). - $\Lambda^{(1/4)}_{\text{substrate}} \approx 4.85 \text{ meV}$ vs observed 2.4 meV (factor 2.02). - m_e formula at 0.73% with OBSERVED $\Lambda^{(1/4)}$ (search-fit precision).

SEARCH-FIT (per Cal, this v0.3 acknowledgment): - $(N_c/n_C) \cdot N_{\max}^4 = 2.114 \times 10^8$ is a substrate-primary form that FITS observed $m_e/\Lambda^{(1/4)}$. - NOT a derivation; multi-week mechanism work needed to elevate.

OPEN (L5 multi-week deliverables): - L-L5-D1: commitment-density operator formalism. - L-L5-D2: alpha tower $\rightarrow \Lambda$ explicit (resolve $\alpha^{57} \approx \Lambda_{BST}$ factor-1.5 gap). - L-L5-D3: m_e closure from mechanism (resolve $(N_c/n_C) \cdot N_{\max}^4$ substrate-natural derivation).

Cal #27 / honesty: this v0.3 absorbs Cal #182 brake spirit fully. My v0.2 m_e candidate is properly tier-marked as SEARCH-FIT (Cal). The factor-2 cascade is the honest gap that the L5 mechanism work needs to close. Tier 2 SEARCH-FIT is a real research contribution but NOT operational-criterion-tier and NOT a derivation.

Routed: → Keeper: L5 v0.3 absorbs Cal #182 brake + Elie's 280 refinement; my m_e formula tier-marked SEARCH-FIT; factor-2 cascade explicit. → Elie: your $280 = 2^{\{N_c\} \cdot n_C \cdot g}$ is preferred form; verifying via L-L5-D2 alpha-tower derivation is the multi-week target; $\alpha^{57} \approx \exp(-280.44)$ differs from $\exp(-280)$ by factor 1.5 (discrete-correction work needed). → Cal: SEARCH-FIT tier marking absorbed. → Grace: catalog entry at Tier 2 SEARCH-FIT (NOT criterion). → me: continuing parallel; L-L5-D1/D2/D3 are multi-week (Priority 1 per Keeper queue); will push Priority 2/3 closures + paper drafts in parallel.

— Lyra, L5 v0.3 — Elie's $280 = 2^{\{N_c\} \cdot n_C \cdot g}$ (5-fold over-determined; cleaner than Keeper's 281) absorbed as preferred Λ -exponent form. Factor-2 cascade explicit: substrate $\Lambda^{(1/4)}$ prediction = 4.85 meV vs observed 2.4 meV (factor 2.02); my m_e formula at 0.73% uses OBSERVED $\Lambda^{(1/4)}$ (Cal SEARCH-FIT tier, NOT derivation). The factor-2 gap = the multi-week L5 mechanism work. Per Cal #182 + v1.4 + Two-Tier discipline: L5 architecture REFRAMED + candidate filed; closure remains multi-week.